

Building a pan flute

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August 2026

1 Theoretical part

Now, we're going to put our understanding of acoustics to the test to build our very own instrument; a pan flute!



Figure 1: A pan flute.

As can be seen in Fig. 1, a pan flute consists of many individual pipes. Each pipe will produce a specific tone when you blow over it.

Question: Try to explain in your own words why this works. What happens if you blow over a pipe?

Question: Have a closer look at Fig. 1. Do you notice something interesting about the lengths of the pipes? Do they get longer linearly? Logarithmically? Exponentially? Do you have an idea why? Remember what we learned about tuning systems.

After learning about different tuning systems, we know which frequencies we need to include to be able to play in a certain key. Before we get into this, recall that in our discussion of waves, we talked about harmonic waves in pipes with one and two fixed ends. Recall that the conditions for the occurring wavelengths were

$$l = n \cdot \frac{\lambda}{2} \tag{1}$$

for the n -th mode in a medium with two fixed ends, and

$$l = (2n + 1) \cdot \frac{\lambda}{4} \tag{2}$$

For the n -th mode for a medium with one fixed and one free end.

How does this translate to the case of the pipes? Having two closed off ends for a pipe seems very impractical, because there's no way to blow over it and excite an oscillation. In fact, we must think carefully about what fixed and free ends even mean in the context of sound, which are pressure waves.

Question: At the open end of the pipe, can air flow in and out? Can the pressure change? What about the closed end? Can the air flow in and out? Can the pressure change? How does this correspond to free and fixed ends?

Question: Assuming we want our lowest note to be as low as possible, with a finite amount of pipes available to us, which should we choose? Pipes with two open ends or pipes with one open and one closed end? Why?

Question: How long would the pipe have to be to produce the middle A, so $f = 440$ Hz? What about a pipe with two open ends?

Question: Using equal temperament, calculate the pipe lengths for the chromatic scale.

Answer:

Note	Length (cm)	Note	Length (cm)	Note	Length (cm)
A_4	19.3	A_5	9.7	A_6	4.8
$A_4^\sharp$	18.2	$A_5^\sharp$	9.1	$A_6^\sharp$	4.6
B_4	17.2	B_5	8.6	B_6	4.3
C_5	16.2	C_6	8.1	C_7	4.1
$C_5^\sharp$	15.3	$C_6^\sharp$	7.7	$C_7^\sharp$	3.8
D_5	14.5	D_6	7.2	D_7	3.6
$D_5^\sharp$	13.7	$D_6^\sharp$	6.8	$D_7^\sharp$	3.4
E_5	12.9	E_6	6.4	E_7	3.2
F_5	12.2	F_6	6.1	F_7	3.0
$F_5^\sharp$	11.5	$F_6^\sharp$	5.7	$F_7^\sharp$	2.9
G_5	10.8	G_6	5.4	G_7	2.7
$G_5^\sharp$	10.2	$G_6^\sharp$	5.1	$G_7^\sharp$	2.6

Table 1: Pipe lengths corresponding to different notes.

Optional: If you prefer, you can also do this for Just tuning or Pythagorean tuning. You'll have to pick the key you want to play in first. I suggest you use the lowest note as the key the scale is in. Have a look at Tab. 1 and chose a note that has a reasonable pipe length. Then do the calculations based on that length.

To save material, we'll only build one octave and only the diatonic scale. (So only 8 pipes in total). Pick a key to play in and choose the corresponding notes and pipe lengths from Tab. 1. Now that we've calculated the pipe lengths, we're all set to actually go ahead and build the pipes!

2 Experimental part

If at any point you're unsure about the instructions, feel free to ask the course leaders. Be careful around the saws and the hot glue.

- Make sure you have all the material you'll need. You need: a pipe (you can calculate the total length from your previous considerations), a measuring stick, a hot glue gun, sanding paper, a saw, a pencil, some clay to close off the pipes, tape, your calculations. You can share the material with the other participants.
- Measure your calculated pipe lengths with the measuring stick and mark the lengths with the pencil
- Show the course leaders your measurements and reasoning. **Do this before you start sawing the pipe! If you cut incorrectly, we might not have enough pipes for everyone.**
- Saw the pipes apart where you marked them with the pencil.
- Sand the ends of the pipes with the sanding paper. Make sure there are no splinters sticking out. Those could hurt you while playing the instrument.
- Block one end of each pipe with the clay. Blow over the pipe to check that it produces the desired note. Adjust the clay until the note is correct. Then let the clay dry.

- Arrange the pipes from largest to smallest. Make sure the open ends of the pipes align. This will make it easier to play. When you're satisfied with your arrangement, tape the pipes together provisorically.
- Now, glue the pipes together with the glue gun. Be careful, its tip is very hot! Try not to get any hot glue on your hands and your clothes, and make sure to put a piece of cardboard under your flute before starting to glue. It's quite hard to get the glue off the table again.
- Wait for the glue to cool off. Flip the pipe, and glue it together on the other side, too.
- Remove the tape.
- Congratulations, you now have a functional pan flute!